NOVA SCHOOL OF SCIENCE & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE

GONÇALO GOMES RODRIGUES

Master in Software Engineering Degree

USE OF DIFFERENTIAL PRIVACY FOR UNOBSERVABLE PRIVACY-PRESERVED COMMUNICATION

SOME THOUGHTS ON THE LIFE, THE UNIVERSE, AND EVERYTHING ELSE

MASTER IN COMPUTER SCIENCE AND ENGINEERING

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MASTER IN COMPUTER SCIENCE AND ENGINEERING

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Use of Differential Privacy for Unobservable Privacy-Preserved Communication
Some thoughts on the Life, the Universe, and Everything Else

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Dedicatory lorem ipsum.

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However, without any intention of conditioning the form or content of this text, I would like to add that it usually starts with academic thanks (instructors, etc.); then institutional thanks (Research Center, Department, Faculty, University, FCT / MEC scholarships, etc.) and, finally, the personal ones (friends, family, etc.).

But I insist that there are no fixed rules for this text, and it must, above all, express what the author feels.

"You cannot teach a man anything; you can only help him discover it in himself."

— **Galileo**, Somewhere in a book or speach
(Astronomer, physicist and engineer)

Abstract

We connect to the internet daily, gaining access to an overwhelming flow of information about the world. Freedom of speech is an essential right, assumed as an unquestionable part of our daily lives. However, this right is denied by authoritarian regimes and entities that control the flow of information, in order to maintain power and control over their citizens. These forces have continuously advanced their methods and resources to enforce such oppressive policies more effectively. Therefore, it is crucial to ensure that those living under such oppression gain access to the tools and content denied to them and have the means to communicate freely, securely, and without fear.

To tackle this challenge, we aim to develop a communication network, on top of Tor network, capable of countering state-of-the-art correlation attacks to ensure user anonymity and privacy. By incorporating Differential Privacy principles, the solution introduces carefully calibrated noise into traffic flow and ensures robust and formally proven privacy guarantees, distinguishing itself from solutions that rely on practical testing. Furthermore, Differential Privacy also provides customizable and adjustable privacy controls, empowering users to balance the trade-off between privacy and utility to their needs.

Keywords: Anonymity Networks, Differential Privacy, Privacy-Preserving Communication, Covert Communication, Traffic Analysis Attacks

Resumo

Ligamo-nos diariamente à Internet, tendo acesso a um fluxo avassalador de informações sobre o mundo. A liberdade de expressão é um direito essencial, assumido como parte inquestionável do nosso quotidiano. No entanto, este direito é negado por regimes autoritários e entidades que controlam o fluxo de informação, a fim de manter o poder e o controlo sobre os seus cidadãos. Estas forças têm vindo a desenvolver continuamente os seus métodos e recursos para aplicar essas políticas opressivas de forma mais eficaz. Por essa razão, é crucial garantir que aqueles que vivem sob tal opressão tenham acesso às ferramentas e conteúdos que lhes são negados e disponham dos meios para comunicar livremente, em segurança e sem medo.

Para fazer face a este desafio, pretendemos desenvolver uma rede de comunicação, sobre a rede Tor, capaz de contrariar ataques de correlação de última geração para garantir o anonimato e a privacidade dos utilizadores. Ao incorporar os princípios da Privacidade Diferencial, a solução introduz ruído cuidadosamente calibrado no fluxo de tráfego e assegura garantias de privacidade robustas e formalmente comprovadas, distinguindo-se das soluções que se baseiam em testes práticos. Além disso, a Privacidade Diferencial também fornece controlos de privacidade personalizáveis e ajustáveis, permitindo que os utilizadores equilibrem o compromisso entre privacidade e utilidade de acordo com as suas necessidades.

Palavras-chave: Redes de Anonimato, Privacidade Diferencial, Comunicação Preservadora de Privacidade, Comunicação Encoberta, Ataques de Análise de Tráfego

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1

Introduction

The internet has become an integral part of the daily life for most of the world's population. However, in some regions, individuals are denied the ability to freely share their stories or access vital information due to oppressive conditions and restrictions on freedom of speech. This oppression and restrictions are often enforced by totalitarian regimes that have vast resources and power to control the flow of information. [2, 35, 1]

As the demand for anonymity networks continues to expand, so does the necessity for robust privacy guarantees to safeguard secure and confidential communications. At the same time, authoritarian regimes intensify their efforts to restrict access exclusively to state-approved content, employing advanced measures to enforce their censorship policies and maintain control over information flow.

In order to enforce and control the network traffic, these regimes resort to the use of traffic analysis techniques to passively monitor the traffic and infer information about the users. Traffic analysis attacks are a significant threat to anonymity networks, as they can be used to infer users' activities and deny their privacy, as well as deanonymize them. [4, 32, 14] Some have demonstrated that advances in machine learning lead to more sophisticated and effective traffic correlation attacks, such as DeepCorr [19] and DeepCoFFEA [22], which leverage deep learning techniques to correlate traffic patterns and infer users' activities with high accuracy, in anonymity networks like Tor. [15]. Statistical Disclosure attacks also pose a threat to anonymity networks, as they identify sender-receiver pairs in anonymity networks. [8, 18] Website Fingerprinting attacks represent a major threat to anonymity networks, as they allow attackers to learn information about user's online activities, even on encrypted networks like Tor, by eavesdropping and collecting side-channel information from the network traffic, between the user and the entry node. With this information, the attacker can use machine learning techniques to infer the user's online activities, such as the websites visited, by analyzing the traffic patterns, with high accuracy. [27, 24, 6]

As these attacks become more sophisticated and effective, journalists, whistleblowers and any other person trying to express their freedom of speech and access information in these regions, are at risk of being identified and persecuted, leading to the decrease of this networks utility, mistrust and fear. Therefore, it is critical that anonymity networks

become more resilient to these attacks and provide stronger privacy guarantees to their users.

1.1 Motivation

Tor, one of the most popular anonymity networks, is an anonymous communication service based on the Onion Routing Protocol. [9] Tor is designed to protect users' privacy and security by routing their traffic through a network of volunteer-operated servers, encrypting it at each step, and ensuring that no single server knows both the source and destination of the traffic.

Anonymity networks remain vulnerable to attacks such as traffic analysis, which threatens the privacy and security of the users these networks are designed to protect. [4, 32, 14, 8, 18, 27, 24, 6] To address this challenge of circumventing censorship, researchers have proposed a range of solutions. Among these solutions, we discuss traffic encapsulation, fully encrypted protocols, and traffic shaping techniques.

Traffic encapsulation is a method that conceals data by embedding packets within other data formats or protocols, thereby obfuscating their true content. This technique leverages protocols, such as those for web browsing and video streaming, to disguise network traffic, making it indistinguishable. [28, 31, 3, 10, 13]

Traffic Shaping is another technique that aims to regulate data transmission to optimize network performance and to prevent timing and correlation attacks, contributing for privacy-prevention. Similarly, traffic shuffling is a technique that focus on reordering packets, with statistical or cryptographic techniques, to make it difficult for an attacker to correlate the traffic. [25, 16, 7, 23, 33]

Researchers have proposed various solutions. Hopper and Vasserman demonstrated that K-Anonymity is effective against traffic analysis attacks, such as Statistical Disclosure and long-term intersection attacks. [12] Nunes et al. proposed BriK, a Tor pluggable transport, that ensures uniform traffic patterns to make users indistinguishable. [21] Some work as Walkie-Talkie [29], Supersequence [30] and Glove [20] also use traffic shaping for defending against website fingerprinting attacks. Wright, Coull, and Monrose proposed Traffic Morphing [34], a traffic shaping technique that uses a Markov Chain to generate traffic patterns that are indistinguishable from the original traffic.

1.2 Problem

Although anonymity networks are designed to provide robust privacy and security, advancements in traffic analysis and correlation techniques have increasingly exposed vulnerabilities, reducing their reliability and trustworthiness for users—particularly in regions where pervasive surveillance by authoritarian regimes is a critical concern. These regimes actively invest in sophisticated surveillance methods and traffic manipulation strategies to maintain control over digital communication.

Tor, despite being one of the most widely used anonymity networks, is not immune to these challenges. It remains vulnerable to several attack vectors, including traffic correlation, congestion-based attacks, timing-based correlations, and fingerprinting techniques. [4, 15, 32, 14] Such vulnerabilities highlight the ongoing arms race between those striving to safeguard online anonymity and those seeking to undermine it through technical and infrastructural advancements.

The absence of standardized privacy metrics for evaluating privacy guarantees in a formal and rigorous manner raises concerns about the effectiveness of anonymizing networks like Tor. Research has demonstrated vulnerabilities in the Tor network to various attacks, including traffic analysis. For instance, Chakravarty et al. showed that statistical correlation techniques can be used to perform successful traffic analysis attacks on the Tor network. Additionally, Shi and Matsuura demonstrated that user privacy can be compromised through fingerprinting attacks.

1.3 Goals

The goal is to add differential privacy (DP) mechanisms into the Tor open-source project to preserve its existing security guarantees while enhancing its privacy protections. By incorporating DP, the aim is to provide quantifiable and adjustable privacy guarantees to users, addressing vulnerabilities to correlation attacks and improving resistance to traffic observability. Furthermore, the project seeks to analyze the trade-offs between privacy and utility within the Tor context and to evaluate the performance of the proposed solution compared to the original Tor network.

As stated in Section, it is important to ensure that anonymity networks are secure against threats like artificial intelligence and machine learning, and to address the lack of standardized privacy metrics to evaluate the performance of anonymity networks. We aim to create and implement a new algorithm that can be used in anonymity networks that leverages differential privacy to ensure privacy and security with higher performance than existing solutions. We also aim to evaluate the performance of the algorithm and compare it with other anonymity networks to prove that this solution is a viable alternative to usual anonymity networks solutions. To test the algorithm and its performance, we intend to realize correlation attacks on the existing networks and on our prototype and compare the results. To test performance, we will measure the latency and throughput of the algorithm and compare it with other anonymity networks.

1.3.1 Contributions

This thesis tries to address the problem of ensuring privacy and security in anonymity networks while maintaining high performance. In particular, the contributions of this thesis are:

1. Design a new Differential Privacy Based Algorithm for Privacy Preserved Communication in Anonymity Networks.
2. Implement the algorithm on top of the Tor network and provide a proof of concept.
3. Evaluate the performance of the algorithm and compare it with other anonymity networks; This contribution aims to prove if this solution is a viable alternative to usual anonymity networks solutions and if, as mention before, can allow users to switch from traditional to anonymity networks.

1.4 Report Organization

The remainder of this report is organized as follows: Chapter introduces anonymity networks and differential privacy, as well as some other related topics. The proposal of work is presented on Chapter, which provides some system model and guidelines for the development and discusses the proof method to verify the work. Finally, workplan is explained and addressed in Chapter

2

Related Work

2.1 Tor Network

Tor is a circuit-based low-latency anonymous communication service based on *onion routing* for asynchronous, loosely federated onion routers [Tor: The second-generation onion router]. This protocol is a low-latency anonymity network design that tries to protect users from correlation attacks. The overlay network relies on *Onion Routers* (ORs) that maintain TLS connection to every other onion router [9].

Onion Proxy are local software, run on the client and handle connections with the users' applications, fetch the current network information and the list of *Onion Routers* (ORs) and build circuits through the network. Each router maintains TLS connections to nodes they have been in contact recently and a pair of keys: long-term identity key and short-term onion key. Long-term keys are used to sign TLS certificates and its router descriptor. Short-term keys are used to decrypt requests, negotiate ephemeral keys and set up circuits.

Traffic travels the network through 512 bytes fixed-size cells with a header and a payload. The header contains the circuit identifier, the command to describe what to do with the payload. Based on the header's command, cells are either control cells or relay cells. Control cells are used to manage circuits and connections, while relay cells are used to carry data.

The sequences of ORs which traffic is routed through are called *circuits*. The user's Onion Proxy construct circuits, negotiating a symmetric key with each OR on the circuit, one hop at a time. Circuits incrementally extends, as well as torn down. Relay cells can only be sent after a circuit is established. Once established, relay cells are encapsulated in several layers of encryption, correspondent to each node in the circuit, with the pre-negotiated keys with each router. As it passes through each node, a layer of encryption is peeled off and forward to the next node. This ensures that only the last node knows the final destination of the data, and no single node knows both the source and destination of the data.

2.2 Anonymity Networks & Mixnets

Mix networks (Mixnets) were introduced by Chaum[5] as a public key cryptography based protocol designed to obscure the relationship between senders and receivers in a communication network, as well as its content, without the need for a universal trusted authority. Mixnets consists of a series of intermediary servers, called mixes, through which messages are decrypted, encrypted, randomly permuted and sent forward. To transmit messages, participants encapsulate their data within successive layers of encryption, with each layer corresponding to a specific mix in the network—a method conceptually akin to onion routing [11]. The length of the encrypted messages is proportional to the number of mixes. Upon receiving a message, each mix removes the outermost layer of encryption, processes the decrypted instructions, and forwards the partially decrypted message to the subsequent mix. This iterative procedure continues until the message arrives at the final mix, which removes the last layer of encryption and transmits the plaintext message to the intended recipient. By design, this system ensures that no single mix possesses complete knowledge of both the sender's identity and the recipient's address, thereby achieving a robust level of anonymity.

Onion Routing [11] followed a similar approach, but focused on low latency communication.

2.3 K-anonymity

2.4 TorKameleon & Stream Covered Channels

2.5 Differential Privacy

2.6 Summary

3

Elaboration

3.1 System Model

3.2 Prototype Guidelines

3.3 Validation

3.4 Summary

4

Workplan

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